The Carryover Recursion in the Published pmsimstats Code: Compounding Decay of the Off-Drug Drug-Response Mean

2026-10-08 15:21 PDT

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Purpose. This note describes one specific feature of the code that accompanies Hendrickson et al. (2020) [1], the `pmsimstats` package at commit `58b32a9` [2]. When a simulated participant stops the drug, the code is meant to let the expected drug response decay exponentially from its last on-drug value with a stated half-life, as the article describes. The code instead decays the previous visit's mean, which has already been decayed, by the cumulative time since discontinuation. The decay therefore compounds from the second off-drug visit of a run onward and runs faster than the stated half-life. The note quotes the code verbatim, derives the size of the discrepancy, works two examples with every number shown, states what the discrepancy does and does not change downstream, and gives a one-line correction with a check that it reproduces the intended equation exactly. It is written to be read on its own, by readers who have not seen any other pmsimstats-ng document, and in particular by the authors of the article and the original code, who are best placed to confirm or correct the reading given here.

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Notation index and glossary

Notation follows the project's canonical table (`analysis/report/NOTATION.md`). Symbols marked *local* are defined here for this note only.

Table 1: Notation index for this note

Symbol	Meaning	Status
t	visit index within a randomization path	canonical
D_{it}	binary drug state: 1 on drug, 0 off drug	canonical
t_{od}	cumulative time on drug at a visit (code `tod`); the input to the Gompertz curve	canonical
t_{sd}	time since discontinuation (code `tsd`), cumulative within an off-drug run, zero on drug and before first exposure	canonical
$t_{1/2}$	carryover half-life, in weeks (code `carryover_t1half`)	canonical
λ	decay rate, $\ln 2/t_{1/2}$	canonical
BR	the biological (drug) response component	canonical
$G(t_{od})$	modified Gompertz mean of BR on drug (code `modgompertz`)	proposed
$\mu_{BR,t}$	expected BR at visit t (code `brmeans[p]`)	proposed
μ_{last}	μ_{BR} at the last on-drug visit before an off-drug run	local
m	position of an off-drug visit within its run, $m = 1, \ldots, L$	local
L	number of off-drug visits in a run	local
j	running index over the off-drug visits of a run, $j \leq m$	local
$t_{sd,m}$	cumulative t_{sd} at the m-th off-drug visit of a run; $t_{sd,0} = 0$	local
μ_m	μ_{BR} at the m-th off-drug visit of a run; $\mu_0 = \mu_{\text{last}}$	local
r_m	ratio of the published to the intended mean at the m-th off-drug visit	local
$t_{1/2}^{\text{eff}}(m)$	effective half-life of the published recursion at the m-th off-drug visit	local
Δt	visit spacing within an off-drug run, when equal, in weeks	local
s_f	the scale factor added to the exponent in 2024 (code `scalefactor`)	local
c_{bm}	on-drug correlation of the biomarker with BR (code `c.bm`)	canonical
$\widehat{\beta}_{bm:D}$	estimated biomarker-by-treatment interaction	canonical
N, α	total sample size; nominal significance level	canonical
$D_{bc,it}$	continuous exposure-decayed drug indicator (code `Dbc`)	canonical
μ_p, $\bar{\mu}$	path mean and its average, in the between-path term of `docs/37`	local (`docs/37`)

Glossary.

- **Hybrid design.** The design that the article calls the "N-of-1 design": eight visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, with open-label titration, a blinded discontinuation and a brief crossover, and four randomization paths (`pathA` to `pathD`, here paths 1 to 4).
- **OL+BDC.** Open-label treatment followed by blinded discontinuation; two paths.
- **CO.** Two-period crossover, eight visits 2.5 weeks apart; two paths.
- **Off-drug run.** A maximal sequence of consecutive off-drug visits after the participant has been exposed to the drug.
- **Anchored decay.** Decay of the off-drug mean from μ_{last} by the elapsed time t_{sd}; the model the article describes.
- **Compounding decay.** Decay applied to an already-decayed value with the cumulative elapsed time, so that elapsed time is counted more than once.
- **The published process.** The data-generating process of the `58b32a9` code (`NOTATION.md`); used here only where a name is needed.
- **The refined process.** The pmsimstats-ng data-generating process, which among other changes uses anchored decay (`NOTATION.md`).

Evidence labels. Each claim carries one of four labels: *quoted* (text reproduced from a source), *inspected* (read from source code), *derived* (follows algebraically) or *verified* (computed by running code, with the script named in Section 12).

1. Summary for the busy reader

What the code does. In `R/generateData.R` at `58b32a9`, line 88 sets the expected drug response at an off-drug visit to the previous visit's expected response times $2^{-t_{sd}/t_{1/2}}$, where t_{sd} is the *cumulative* time since the drug was stopped (inspected). Because the previous visit's value has already been decayed, elapsed time is counted again at every visit, and at the m-th off-drug visit the mean is $\mu_{\text{last}}\, 2^{-(t_{sd,1}+\cdots+t_{sd,m})/t_{1/2}}$ (derived).

What it should do. The article states that "the value of the BR at the last timepoint on active drug was exponentially decayed, with the half-life of this decay being maintained as a model parameter" [1, p. 6] (quoted). That is $\mu_{\text{last}}\, 2^{-t_{sd,m}/t_{1/2}}$.

Where the two agree and where they differ. They agree at every on-drug visit and at the first off-drug visit of every run. From the second off-drug visit on, the code's mean is smaller by the factor $2^{-(t_{sd,1}+\cdots+t_{sd,m-1})/t_{1/2}}$ (derived). With equally spaced visits the effective half-life at the m-th visit is $2t_{1/2}/(m+1)$: two-thirds of the stated value at the second visit, one-half at the third (derived).

How large. On Hybrid path 1 at $t_{1/2} = 1$ week, the mean two weeks after stopping is 1.273 points where the intended value is 2.547 (verified, Section 6). The largest shortfall in any off-drug mean over the three designs' paths is 1.37 points at $t_{1/2} = 1$ week and 2.50 points at 2 weeks (verified, Table 5). At the half-lives used for the article's power results, 0.1 and 0.2 weeks, the entire carryover is nearly gone after one week under either form, and the largest shortfall is 0.0104 points (verified). The discrepancy is therefore numerically negligible for those results, and grows with the half-life and with the length of the off-drug run.

What it does not touch. The variances and the correlation matrix are unchanged, so the simulated data differ only by a shift in the mean at the affected visits (inspected, verified). A paired simulation with the published analysis shows no detectable change in power from correcting the mean alone (verified, Section 8.3).

The one-line fix. Decay by the increment in t_{sd} since the previous visit rather than by its cumulative value:

```
brmeans[p]<-brmeans[p]+brmeans[p-1]*
  (1/2)^((d$tsd[p]-d$tsd[p-1])/modelparam$carryover_t1half)
```

The product of the increments telescopes to the intended $2^{-t_{sd,m}/t_{1/2}}$; the corrected code reproduces the intended means to within 1.1×10^{-14} in all 40 path and half-life cases checked (verified, Section 9).

2. Scope

This note concerns one statement: the off-drug carryover of the mean of the drug-response component. It does not address the biomarker coupling rule, the correlation structure, positive-definiteness repair or the analysis model, each of which is examined elsewhere in the project. Where those features interact with the carryover of the mean, the interaction is stated in Section 8.

3. The intended model

The article describes the drug-response component as follows (quoted; [1], Methods, p. 6):

> The BR factor was set to zero at times when participants had never been on active drug; however, a carryover effect was built in such that when a participant moved from active drug to being off active drug, the value of the BR at the last timepoint on active drug was exponentially decayed, with the half-life of this decay being maintained as a model parameter.

and, in the Discussion (quoted; [1], p. 12):

> In these models, we assume an exponential decay analogous to a pharmacokinetic half-life, although the simulations could be easily adapted to incorporate an alternative model.

In words: when the drug is stopped, the expected drug response starts at its last on-drug value and halves every $t_{1/2}$ weeks. Writing μ_{last} for that value and $t_{sd,t}$ for the time since discontinuation at visit t,

$$\mu_{BR,t} \;=\; \mu_{\text{last}}\, 2^{-t_{sd,t}/t_{1/2}} \;=\; \mu_{\text{last}}\, e^{-\lambda\, t_{sd,t}}, \qquad \lambda = \ln 2 / t_{1/2}.$$

This is the anchored form. It is a single exponential in elapsed time, so its half-life is $t_{1/2}$ at every visit, and it does not depend on how many visits fall in the off-drug run. Carryover in within-subject designs has a long methodological literature [4]; the exponential form is a pharmacologically motivated choice within it, and it is the one the article's text describes.

4. The code as published

4.1 The carryover lines

`R/generateData.R` at `58b32a9`, lines 83 to 92 (quoted; permalink):

```
83      brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
84      if(nP>1){
85        for(p in 2:nP){
86          if(!d[p]$onDrug){
87            if(d[p]$tsd>0){
88              brmeans[p]<-brmeans[p]+brmeans[p-1]*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
89            }
90          }
91        }
92      }
```

The drug state used on line 86 is set at line 55 (permalink):

```
53  d<-data.table(trialdesign)
54  d$t_wk_cumulative<-cumulative(d$t_wk)
55  d[,onDrug:=(tod>0)]
```

The file `R/generateData.R` was last changed before `58b32a9` in the initial commit `42ac030` (2020-02-22), so these lines are the ones that produced the article's results (inspected, GitHub commit history for the file).

4.2 How the trial design supplies t_{od} and t_{sd}

`R/buildtrialdesign.R` at `58b32a9`, lines 86 to 106 (quoted; permalink):

```
 86  t_wk<-c(timepoints[1],
 87          timepoints[2:length(timepoints)]-timepoints[1:(length(timepoints)-1)])
 88  for(iTD in 1:nTD){
 89    od<-ondrug[[iTD]]
 90    od_intervals<-t_wk*od
 91    tod<-od_intervals
 92    tsd<-t_wk*(od!=1)
 93    tpb<-t_wk*(expectancies>0)
 94    for(iTP in 2:length(timepoints)){
 95      if(od_intervals[iTP]>0){
 96        tod[iTP]<-tod[iTP]+tod[iTP-1]
 97      }else{
 98        tsd[iTP]<-tsd[iTP]+tsd[iTP-1]
 99      }
100      if(tpb[iTP]>0){
101        tpb[iTP]<-tpb[iTP]+tpb[iTP-1]
102      }
103    }
104    everondrug<-(cumulative(od)>0)
105    tsd<-everondrug*tsd
106    tds[[iTD]]<-data.table(timeptnames=timeptnames,t_wk=t_wk,e=expectancies,tod=tod,tsd=tsd,tpb=tpb)
```

The modified Gompertz curve, `R/utilities.R` lines 34 to 40 (quoted; permalink):

```
 34  modgompertz<-function(t,maxr,disp,rate){
 35    y<-maxr*exp(-disp*exp(-rate*t))
 36    vert_offset<-maxr*exp(-disp*exp(-rate*0))
 37    y<-y-vert_offset
 38    y<-y*(maxr/(maxr-vert_offset)) # return the assymptotic max to what it was
 39    return(y)
 40  }
```

4.3 Line-by-line reading

All statements in this subsection are inspected.

- **`buildtrialdesign`, lines 86 to 87.** `t_wk` holds the interval lengths: the first visit's week, then the gaps between visits.
- **Lines 90 to 92.** Each interval is assigned to time on drug if the participant is on drug at its closing visit (`od = 1`, "on drug entire previous interval and this measurement point", in the package's own comment), and to time off drug otherwise.
- **Lines 94 to 99.** Moving forward through the visits, an on-drug interval adds to the previous `tod`; an off-drug interval adds to the previous `tsd`. Two consequences follow. First, `tsd` is *cumulative within an off-drug run*: one week, then two, then three. Second, `tsd` returns to 0 at every on-drug visit, and `tod` is 0 at every off-drug visit, so time on drug restarts at zero on re-exposure.
- **Lines 104 to 105.** `tsd` is set to zero before first exposure, so a participant who starts on placebo has no carryover.
- **`generateData`, line 83.** `brmeans` is the Gompertz mean $G(t_{od})$ at every visit. Because $G(0) = 0$ (line 37 subtracts the value at zero), it is 0 at every off-drug visit.

- **Lines 85 to 87.** The loop runs forward from the second visit and acts only at off-drug visits after exposure ($t_{sd} > 0$).
- **Line 88.** The off-drug mean becomes $0 +$ `brmeans[p-1]` $\times\, 2^{-t_{sd}/t_{1/2}}$. Since the loop runs forward, `brmeans[p-1]` has already been overwritten at the previous iteration whenever visit $p-1$ was itself off drug. At the first off-drug visit of a run it is μ_{last}; at later visits it is the previous, already-decayed value. The exponent, however, uses the cumulative `d$tsd[p]`.

The pairing on line 88 of an updated (recursive) value with a cumulative (non-recursive) elapsed time is the source of the discrepancy. Either element alone would be correct: a recursion with the one-interval increment, or the cumulative time applied to μ_{last}.

4.4 What the 2024 revisions changed

Three later commits changed `R/generateData.R` (inspected, GitHub commit history and per-commit patches):

- **`8609f12` (2024-05-06).** Adds an argument `scalefactor = 2` to the signature (line 52 of the current file) and multiplies the exponent on the carryover line by it (line 96 of the current file, permalink):

  ```
  brmeans[p]<-brmeans[p]+brmeans[p-1]*(1/2)^(scalefactor * d$tsd[p]/modelparam$carryover_t1half)
  ```

  The argument is documented as `TODO update when understand what this does?` (line 43). The same commit, whose message credits edits by Ron Thomas, also changes the biomarker correlation rule. That rule is the subject of the companion note `docs/55`, but it depends on the carryover recursion in a way that matters for the fix (Section 9.5).

- **`f6ee86d` (2024-05-08) and `325314f` (2024-05-10).** Correct an index in the same-visit cross-factor correlation (`correlations[n2,n1]`) and adjust its spacing. Neither touches the carryover lines.

The current upstream code (`master` at `06dac83`, 2024-11-27) therefore keeps the recursion with cumulative t_{sd} and adds the factor $s_f = 2$ to the exponent. Its driver `generateSimulatedResults()` calls `generateData()` without the argument (lines 138 and 144), so the default $s_f = 2$ applies (inspected). At the m-th off-drug visit of a run the current code gives

$$\mu_{BR} = \mu_{\text{last}}\, 2^{-s_f\,(t_{sd,1}+\cdots+t_{sd,m})/t_{1/2}},$$

which is the published recursion evaluated at half the stated half-life (derived). The compounding remains, and with $s_f = 2$ the first off-drug visit of each run is also below the intended value, by the factor $2^{-t_{sd,1}/t_{1/2}}$ (derived). Table 2 gives the values on Hybrid path 1. The current upstream file was run with `empirical = TRUE` (Section 6.1) in all 40 cases of Section 9 and matched this formula to within 1.6×10^{-14} (verified).

Table 2: Off-drug drug-response means on Hybrid path 1 at $t_{1/2} = 1$ week: the 58b32a9 code, the current upstream code and the intended model (verified)

Visit	Week	t_{sd}	Published, $s_f = 1$	Current, $s_f = 2$	Intended
BD3	11	1	5.094	2.547	5.094
BD4	12	2	1.273	0.159	2.547
COp	20	4	0.268	0.017	0.268

5. Why it compounds

Consider one off-drug run that follows a last on-drug mean μ_{last}, with off-drug visits $m = 1, \ldots, L$ at cumulative times since discontinuation $t_{sd,1} < t_{sd,2} < \cdots < t_{sd,L}$. Line 88 gives, with the Gompertz term zero off drug,

$$\mu_m = \mu_{m-1}\, 2^{-t_{sd,m}/t_{1/2}}, \quad \mu_0 = \mu_{\text{last}} \qquad \Longrightarrow \qquad \mu_m = \mu_{\text{last}}\, 2^{-(t_{sd,1}+t_{sd,2}+\cdots+t_{sd,m})/t_{1/2}},$$

while the intended model gives $\mu_{\text{last}}\, 2^{-t_{sd,m}/t_{1/2}}$ (derived). The ratio of published to intended is

$$r_m = 2^{-(t_{sd,1}+\cdots+t_{sd,m-1})/t_{1/2}},$$

equal to 1 at $m = 1$ and strictly less than 1 for every $m \geq 2$ (derived).

Equal spacing. With visits Δt weeks apart, $t_{sd,j} = j\,\Delta t$, the sum is $\Delta t\, m(m+1)/2$, and

$$\mu_m = \mu_{\text{last}}\, 2^{-\Delta t\, m(m+1)/(2t_{1/2})}, \qquad r_m = 2^{-\Delta t\, m(m-1)/(2t_{1/2})}$$

(derived). Equivalently, the code behaves at the m-th visit like the intended model with an effective half-life

$$t_{1/2}^{\text{eff}}(m) = \frac{t_{sd,m}}{t_{sd,1} + \cdots + t_{sd,m}}\, t_{1/2} \;=\; \frac{2\, t_{1/2}}{m+1} \quad \text{(equal spacing)},$$

so the effective half-life is $t_{1/2}$ at the first visit, $2t_{1/2}/3$ at the second, $t_{1/2}/2$ at the third and $2t_{1/2}/5$ at the fourth (derived). It shrinks with every visit in the run, and the shortfall depends on how densely the off-drug period is sampled: the same elapsed time measured at more visits decays further. The intended model has no such dependence.

6. Worked example 1: Hybrid path 1

6.1 Set-up

The design is the article's Hybrid ("N-of-1") design as defined in the package's publication vignette (`vignettes/Produce_Publication_Results_1_generate_data.Rmd`, lines 48 to 61 at `58b32a9`): visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, path 1 (`pathA`) on drug at visits 1 to 4 and 7, off at visits 5, 6 and 8 (inspected). The drug-response parameters are the shipped `extracted_rp` values for `br`: maximum 10.98604, displacement 5, rate 0.42 (inspected; `data/extracted_rp.rda` at `58b32a9`).

The means in this section were obtained in two independent ways that agree to 10^{-14}: by restating lines 83 to 92 as a function, and by running the published `generateData()` itself with `empirical = TRUE`, for which `MASS::mvrnorm()` returns a sample whose column means equal the supplied mean vector exactly [3], and reading off the column means (verified).

6.2 Step by step

The Gompertz means at the on-drug visits are $G(4) = 4.281307$, $G(8) = 9.222638$, $G(9) = 9.793228$, $G(10) = 10.187065$ and, after the restart at re-exposure, $G(4) = 4.281307$ again at week 16 (verified). At $t_{1/2} = 1$ week:

- **BD3, week 11,** $t_{sd} = 1$**.** Published: $0 + 10.1871 \times 2^{-1/1} = 5.0935$. Intended: the same, 5.0935.

- **BD4, week 12, $t_{sd} = 2$.** Published: the previous value is carried forward and decayed by the cumulative two weeks, $5.0935 \times 2^{-2/1} = 1.2734$, which is $10.1871 \times 2^{-(1+2)/1}$. Intended: $10.1871 \times 2^{-2/1} = 2.5468$. Ratio 0.500.
- **COd, week 16.** On drug; t_{od} restarts at 4 weeks; 4.2813 in both.
- **COp, week 20, $t_{sd} = 4$.** The first visit of a new run, so both give $4.2813 \times 2^{-4/1} = 0.2676$.

At $t_{1/2} = 0.5$ week the same steps give 2.5468 at BD3 in both, then $2.5468 \times 2^{-2/0.5} = 0.1592$ (published) against $10.1871 \times 2^{-2/0.5} = 0.6367$ (intended) at BD4, a ratio of 0.250, and 0.0167 in both at COp (verified).

Table 3: Hybrid path 1: expected drug response by visit under the published recursion (Pub.) and the intended anchored decay (Int.), at $t_{1/2} = 1$ and 0.5 week (verified)

Visit	Week	D_{it}	t_{od}	t_{sd}	$G(t_{od})$	Pub., 1	Int., 1	Ratio, 1	Pub., 0.5	Int., 0.5	Ratio, 0.5
OL1	4	1	4	0	4.281	4.281	4.281		4.281	4.281	
OL2	8	1	8	0	9.223	9.223	9.223		9.223	9.223	
BD1	9	1	9	0	9.793	9.793	9.793		9.793	9.793	
BD2	10	1	10	0	10.187	10.187	10.187		10.187	10.187	
BD3	11	0	0	1	0.000	5.094	5.094	1.000	2.547	2.547	1.000
BD4	12	0	0	2	0.000	1.273	2.547	0.500	0.159	0.637	0.250
COd	16	1	4	0	4.281	4.281	4.281		4.281	4.281	
COp	20	0	0	4	0.000	0.268	0.268	1.000	0.017	0.017	1.000

The ratio column agrees with Section 5: BD4 is the second visit of a run with $\Delta t = 1$, so $r_2 = 2^{-1/t_{1/2}}$, which is 0.5 at $t_{1/2} = 1$ and 0.25 at $t_{1/2} = 0.5$ (derived). The effective half-life at BD4 is $2t_{1/2}/3$: 0.67 week instead of 1, and 0.33 instead of 0.5.

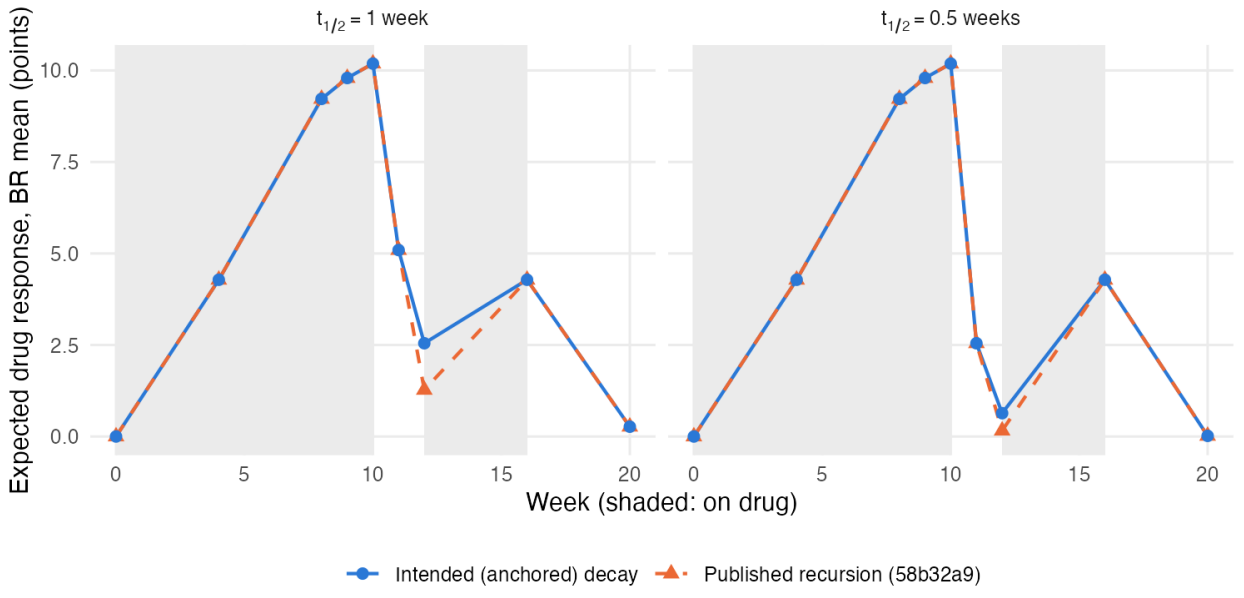


Figure 1: Hybrid path 1: expected drug response under the published recursion (dashed, triangles) and the intended anchored decay (solid, circles), at $t_{1/2} = 1$ week (left) and 0.5 week (right). Points are visit means; the lines join them and do not show the trajectory between visits. Shaded intervals are on drug. Baseline is plotted at 0. The two agree everywhere except at week 12, the second visit of the off-drug run.

7. Worked example 2: the crossover

The crossover design of the vignette (lines 87 to 98) has eight visits 2.5 weeks apart; path 1 (`pathA`) is on drug for the first four and off for the last four (inspected). The off-drug run has four visits at $t_{sd} = 2.5, 5, 7.5, 10$, so by Section 5 the ratios are $r_m = 2^{-2.5\,m(m-1)/(2t_{1/2})}$: at $t_{1/2} = 2$ weeks, $1, 2^{-1.25}, 2^{-3.75}, 2^{-7.5}$, that is 1, 0.420, 0.074 and 0.0055 (derived; verified in Table 4). The effective half-life at the fourth off-drug visit is $2 \times 2/5 = 0.8$ week instead of 2.

Table 4: Crossover path 1: expected drug response by visit under the published recursion (Pub.) and the intended anchored decay (Int.), at $t_{1/2} = 1$ and 2 weeks (verified)

Visit	Week	D_{it}	t_{od}	t_{sd}	$G(t_{od})$	Pub., 1	Int., 1	Ratio, 1	Pub., 2	Int., 2	Ratio, 2
COa1	2.5	1	2.5	0	1.848	1.848	1.848		1.848	1.848	
COa2	5	1	5	0	5.922	5.922	5.922		5.922	5.922	
COa3	7.5	1	7.5	0	8.853	8.853	8.853		8.853	8.853	
COa4	10	1	10	0	10.187	10.187	10.187		10.187	10.187	
COb1	12.5	0	0	2.5	0.000	1.801	1.801	1.000	4.283	4.283	1.000
COb2	15	0	0	5	0.000	0.056	0.318	0.177	0.757	1.801	0.420
COb3	17.5	0	0	7.5	0.000	0.000	0.056	0.006	0.056	0.757	0.074
COb4	20	0	0	10	0.000	0.000	0.010	0.000	0.002	0.318	0.006

At $t_{1/2} = 2$ weeks the intended mean at week 15, five weeks or two and a half half-lives after stopping, is $10.187 \times 2^{-2.5} = 1.801$ points. The published mean there is 0.757, the value the intended model reaches only at week 17.5. At week 17.5 the published mean, 0.056, is already below the intended value at week 20, 0.318 (verified, Table 4).



Figure 2: Crossover path 1: expected drug response under the published recursion (dashed, triangles) and the intended anchored decay (solid, circles), at $t_{1/2} = 1$ week (left) and 2 weeks (right). Points are visit means joined by lines; shading marks the on-drug period. With four off-drug visits in one run, the published mean falls well below the intended one from the second off-drug visit.

8. Consequences

8.1 What changes: the off-drug means

Only the expected drug response at the second and later visits of an off-drug run changes (derived, Section 5; verified for all paths in Table 5). Table 5 gives, for each design and half-life, the smallest published-to-intended ratio and the largest shortfall in points over all off-drug visits of all paths.

Table 5: Published versus intended off-drug drug-response means: smallest ratio and largest shortfall over every off-drug visit of every path, by design and half-life (verified)

Design	$t_{1/2}$ (weeks)	Smallest ratio	Largest shortfall (points)
Hybrid	0.1	8.7×10^{-19}	9.7×10^{-6}
Hybrid	0.2	9.3×10^{-10}	0.0096
Hybrid	0.5	2.4×10^{-4}	0.478
Hybrid	1	0.0156	1.273
Hybrid	2	0.125	2.238
OL+BDC	0.1	9.3×10^{-10}	1.0×10^{-5}
OL+BDC	0.2	3.1×10^{-5}	0.0104
OL+BDC	0.5	0.0156	0.514
OL+BDC	1	0.125	1.370
OL+BDC	2	0.354	2.501
CO	0.1	7.0×10^{-46}	9.0×10^{-15}
CO	0.2	2.6×10^{-23}	3.0×10^{-7}
CO	0.5	9.3×10^{-10}	0.0096
CO	1	3.1×10^{-5}	0.262
CO	2	0.0055	1.044

The two columns tell different stories, and both matter. The ratio can be extreme at short half-lives, but there the intended carryover is itself almost zero after a week: at $t_{1/2} = 0.1$ week the largest intended off-drug mean anywhere is 0.011 points (verified). The shortfall in points is what reaches the data, and it is negligible at the half-lives of the article's power analyses, 0.1 and 0.2 weeks (the `coremodelparams` grid, vignette line 318, inspected). It becomes material, of the order of one to two and a half points on the CAPS scale, at half-lives of one to two weeks, which the article does not report on but which the package accepts.

8.2 What does not change

- **On-drug means**, and the first off-drug visit of every run, are identical (derived).
- **Variances.** The standard deviations are set at lines 69 to 72 and do not depend on the means or the half-life (inspected).
- **The correlation matrix.** The published coupling rule (lines 132 to 138) correlates the biomarker with the drug response at every visit whose mean is nonzero. Both forms are strictly positive at exactly the same visits, so the matrix is the same (inspected; the pattern of nonzero means is identical in all 40 cases, verified). For the same seed, then, the simulated data under the two forms differ only by a fixed shift at the affected visits.
- **The biomarker-by-drug interaction in the generated data.** Under the published process the interaction is carried by the correlation matrix, not by the mean, so the true interaction is unchanged (inspected).

8.3 Effect on the published analysis

Because only the mean moves, any effect on inference runs through how the analysis model absorbs a different off-drug mean profile. Two pieces of evidence bear on this.

A closed-form argument (quoted). For the paired-difference statistic E9, a two-stage proxy for the published mixed model, `docs/37` (Section 4.4) shows that the means enter the power only through a between-path term $(\mu_p - \bar{\mu})^2$, and reports that "In the Hybrid and OL+BDC designs the path-mean term is small: in Hybrid at most 0.30 against residual variances of 41-49, under 0.7% of the total, decreasing with $t_{1/2}$". The compendium's mathematics note reaches the same conclusion and labels the effect on the mixed model "inferred, not tested" (`analysis/report/compendium-mathematics.tex`, section "Anchored versus compounding means in the closed form").

A paired simulation (verified; computed for this note). The published `generateData()` was run in four versions: as published; with only lines 83 to 92 replaced by the anchored form of Section 9; with the graded biomarker coupling used elsewhere in the project (`docs/37`, Appendix A); and with both changes. Each version fed the published `generateSimulatedResults()` and `lme_analysis()` (the `lmer` model with `useDE = FALSE`), with the Hybrid design, the published parameters at $N = 70$, $c_{bm} = 0.25$, no censoring and 500 replicates per cell. All versions used the same seed at a given half-life, so each pair of versions differing only in the carryover of the mean shares identical noise.

Table 6: Power of the published analysis for the interaction, Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, by version of the 58b32a9 data generator (verified; Monte Carlo standard error at most 0.022)

$t_{1/2}$ (weeks)	Published	Anchored only	Graded	Graded and anchored
0	0.628	0.628	0.628	0.628
0.1	0.078	0.078	0.658	0.658
0.2	0.082	0.082	0.670	0.670
1	0.088	0.088	0.486	0.484
2	0.078	0.072	0.344	0.344

Table 7: Replicate-by-replicate effect of the anchored correction on the interaction estimate and on the test decision at $\alpha = 0.05$, same data otherwise (verified)

$t_{1/2}$ (weeks)	Pair	Mean change in $\widehat{\beta}_{bm:D}$	Largest change	Discordant decisions
0.1	anchored vs published	3×10^{-11}	6.8×10^{-9}	0 of 500
0.2	anchored vs published	1.2×10^{-7}	6.0×10^{-6}	0 of 500
1	anchored vs published	1.2×10^{-5}	0.0017	0 of 500
2	anchored vs published	-7.4×10^{-5}	0.0049	3 of 500
1	graded and anchored vs graded	5.3×10^{-6}	0.0017	1 of 500
2	graded and anchored vs graded	-7.3×10^{-5}	0.0042	4 of 500

Three things follow (verified unless stated).

- **At $t_{1/2} = 0$ all four versions coincide**, as they must: there is no carryover to correct, and the shared seed gives identical data.
- **Correcting the mean alone leaves the published analysis essentially unchanged.** At the article's half-lives of 0.1 and 0.2 weeks the interaction estimates move by less than 10^{-5} and no test decision changes. At 1 and 2 weeks the estimates move by at most 0.005, against a mean estimate near -0.03 to -0.13, and at most 4 of 500 decisions change; the corresponding power differences (0 to 0.006) are well inside one Monte Carlo standard error. In the published model the time term and the drug main effect absorb most of the change in the off-drug mean profile (inferred).
- **The power loss under carryover in the published code does not come from this recursion.** With the published coupling rule, power falls from 0.628 to about 0.08 at any positive half-life, with or without the correction. With graded coupling it stays near 0.66 at 0.1 and 0.2 weeks and declines to 0.49 and 0.34 at 1 and 2 weeks, again with or without the correction. The coupling rule, which is a separate feature of the code (`docs/37`, Appendix A, where the same comparison at $t_{1/2} \in \{0, 0.1, 0.2, 1\}$ with different seeds gave published power 0.668, 0.126, 0.092 and 0.094; quoted), governs the power; the recursion governs only the off-drug means.

In short, for the article's analysis and parameter grid the recursion is a discrepancy between the stated and the simulated decay that has no material effect on the reported power. Its practical importance is for users who set longer half-lives, who use the simulated means themselves (for example to plot expected trajectories, as in the article's Figure 3), or who fit analyses that model the decay.

8.4 Where the discrepancy could matter more

The following are inferred, not tested here.

- **Longer half-lives and longer off-drug runs.** The shortfall grows with both (Section 5, Table 5). Designs with several closely spaced off-drug visits, or half-lives of a week or more, are where the published process departs most from its stated parameter.
- **Analyses that model the decay.** An analysis that weights the off-drug visits by $2^{-t_{sd}/t_{1/2}}$ at the stated half-life (the continuous indicator $D_{bc,it}$ used in later work) assumes the anchored form. Data from the published process decay faster than that indicator, so the analysis is mildly misspecified even when its half-life equals the simulation's.
- **Estimating the half-life.** A procedure that estimates $t_{1/2}$ from simulated data would recover something closer to the effective half-life of Section 5 than to the stated one.

9. The fix

9.1 The minimal change (exact for exponential decay)

Because exponential decay is memoryless, the recursion is correct if each step uses the time elapsed since the previous visit. At the first off-drug visit of a run the previous visit is on drug, where `tsd` is 0, so the increment equals the cumulative value and the first step is unchanged. A one-line change against `58b32a9`:

```
--- a/R/generateData.R
+++ b/R/generateData.R
@@ -85,7 +85,7 @@
         for(p in 2:nP){
           if(!d[p]$onDrug){
             if(d[p]$tsd>0){
-              brmeans[p]<-brmeans[p]+brmeans[p-1]*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
+              brmeans[p]<-brmeans[p]+brmeans[p-1]*(1/2)^((d$tsd[p]-d$tsd[p-1])/modelparam$carryover_t1
             }
```

```
      }
    }
```

The product of the increments telescopes, $\prod_{j \le m} 2^{-(t_{sd,j} - t_{sd,j-1})/t_{1/2}} = 2^{-t_{sd,m}/t_{1/2}}$ with $t_{sd,0} = 0$, which is the intended model (derived).

9.2 The anchored form (general)

The one-line change relies on the exponential form. If the decay family is to be varied, as the article anticipates ("the simulations could be easily adapted to incorporate an alternative model"), the decay should be applied once, to the last on-drug value:

```
--- a/R/generateData.R
+++ b/R/generateData.R
@@ -82,11 +82,12 @@
     if(c=="br"){
       brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
       if(nP>1){
-        for(p in 2:nP){
-          if(!d[p]$onDrug){
-            if(d[p]$tsd>0){
-              brmeans[p]<-brmeans[p]+brmeans[p-1]*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
-            }
+        last_on<-0
+        for(p in 1:nP){
+          if(d[p]$onDrug){
+            last_on<-brmeans[p]
+          }else if(d[p]$tsd>0){
+            brmeans[p]<-brmeans[p]+last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
           }
         }
       }
```

This is the form adopted in the pmsimstats-ng package (version 0.3.0, `R/generateData.R`, lines 276 to 291 in `buildSigma()`, added 2026-09-08; inspected). The rest of that package's data-generating process differs from `58b32a9` in other respects, so it is not a clean test of this change on its own; the patched `58b32a9` file is.

9.3 Check

Both patched versions of the `58b32a9` file were run with `empirical = TRUE` on every path of the Hybrid, OL+BDC and CO designs (eight paths) at $t_{1/2} \in \{0.1, 0.2, 0.5, 1, 2\}$, 40 cases in all, and their drug-response means compared with $\mu_{\text{last}}\, 2^{-t_{sd}/t_{1/2}}$ computed directly. The largest absolute difference was 1.2×10^{-14} for the anchored form and 1.1×10^{-14} for the one-line form, floating-point rounding (verified). On Hybrid path 1 at $t_{1/2} = 1$ week both return 5.094, 2.547 and 0.268 at BD3, BD4 and COp, the intended column of Table 3. At $t_{1/2} = 0$ both forms, like the original, give zero carryover, since $2^{-t_{sd}/0} = 0$ in R (inspected).

9.4 Re-exposure is a separate choice

Neither change alters what happens when the drug is restarted. Time on drug restarts at zero (Section 4.3), and the on-drug mean is $G(t_{od})$ with no residual carried in. That is a modeling choice distinct from the

decay discussed here. One interaction is worth noting. Because the anchored form leaves a larger residual at the end of an off-drug run, the residual can exceed the restarted Gompertz value at the first re-exposure visit, producing a downward step in the mean while the participant is back on drug. On Hybrid path 3 (off at weeks 10 to 12, on at week 16) the intended mean at week 12 is 3.462 at $t_{1/2} = 2$ weeks, below the week-16 value of 4.281, but 5.823 at $t_{1/2} = 4$ weeks, above it (verified). The published recursion gives 1.224 and 3.462 at the same two half-lives. A restart from the residual would remove the step; it is a separate question from the one in this note.

9.5 Applying the fix to the current upstream code

The fix in Sections 9.1 and 9.2 is written against `58b32a9`. Applied to the current upstream code it needs one companion change, because commit `8609f12` made the biomarker correlation depend on the recursion. In the current code the correlation between the biomarker and the drug response at an off-drug visit is c_{bm} times the ratio of consecutive drug-response means, $c_{bm}\,\mu_{BR,t}/\mu_{BR,t-1}$ (the block marked `RON THOMAS VERSION`; inspected). Under the compounding recursion that ratio is $2^{-s_f t_{sd,t}/t_{1/2}}$ with the cumulative t_{sd}, so the rule reproduces a coupling that fades with time since discontinuation, at half-life $t_{1/2}/s_f$ (derived). Under anchored means the same ratio is $2^{-s_f(t_{sd,t}-t_{sd,t-1})/t_{1/2}}$, a constant factor per visit: the coupling falls at the first off-drug visit and then stops falling (derived). Table 8 shows the consequence on the crossover's off-drug run.

Table 8: Coupling factor (multiplier of c_{bm}) under the ratio rule of the current upstream code, crossover path 1, $t_{sd} = 2.5, 5, 7.5, 10$ weeks, $t_{1/2} = 1$ week (verified)

Coupling factor at off-drug visit	1	2	3	4
Intended, $2^{-t_{sd}/t_{1/2}}$	0.177	0.031	0.0055	0.0010
Ratio rule, compounding means ($s_f = 1$)	0.177	0.031	0.0055	0.0010
Ratio rule, anchored means ($s_f = 1$)	0.177	0.177	0.177	0.177

The two upstream changes are therefore entangled: the correlation rule is right only while the means compound, and with the default $s_f = 2$ it fades at half the stated half-life. Fixing the recursion alone would leave a biomarker that stays associated with the drug response long after the drug effect has gone. The recommended correction computes the coupling from the time since discontinuation directly, $c_{bm}\,2^{-t_{sd}/t_{1/2}}$ off drug after first exposure and 0 before it, independent of the means (`docs/55`). With that rule the two fixes are independent and can be applied in either order. It also removes a side effect of the ratio rule: because the ratio needs the previous visit, the current code assigns no coupling at visit 1 (the block is guarded by `if (p > 1)`), so the biomarker is uncorrelated with the drug response at the first visit even when it is on drug, in every design (`docs/54`, Section 3.4; inspected). The direct rule assigns c_{bm} there.

10. How to verify independently

The snippet below needs only `data.table`, `MASS` and `corpcor`, the packages the original code already uses, and a checkout of `rchendrickson/pmsimstats` at `58b32a9` as the working directory. It does not use pmsimstats-ng. It runs the original `generateData()` with `empirical = TRUE`, so that the sample means are the code's own mean vector, and prints them beside the intended anchored values.

```
## Run in a checkout of github.com/rchendrickson/pmsimstats at 58b32a9.
library(data.table); library(MASS); library(corpcor)
```

```r
for (f in c('utilities.R', 'buildtrialdesign.R', 'generateData.R'))
  source(file.path('R', f))
load('data/extracted_rp.rda'); load('data/extracted_bp.rda')
td <- buildtrialdesign(
  timepoints = c(4, 8, 9, 10, 11, 12, 16, 20),
  timeptnames = c('OL1','OL2','BD1','BD2','BD3','BD4','COd','COp'),
  expectancies = c(1, 1, .5, .5, .5, .5, .5, .5),
  ondrug = list(pathA = c(1, 1, 1, 1, 0, 0, 1, 0)))$trialpaths[[1]]
t_half <- 1
mp <- data.frame(N = 200, c.bm = 0, carryover_t1half = t_half,
  c.tv = .8, c.pb = .8, c.br = .8, c.cf1t = .2, c.cfct = .1)
## empirical = TRUE makes the sample means equal the code's mean vector
dat <- generateData(mp, extracted_rp, extracted_bp, td,
  empirical = TRUE, makePositiveDefinite = TRUE)
code <- colMeans(dat[, paste0(td$timeptnames, '.br'), with = FALSE])
rp <- extracted_rp[cat == 'br']
g <- modgompertz(td$tod, rp$max, rp$disp, rp$rate)
intended <- g; last <- 0
for (p in seq_along(g)) {
  if (td$tod[p] > 0) last <- g[p]
  else if (td$tsd[p] > 0) intended[p] <- last * 2^(-td$tsd[p] / t_half)
}
print(data.frame(visit = td$timeptnames, tsd = td$tsd,
  code = round(code, 3), intended = round(intended, 3)), row.names = FALSE)
```

Its output, run against the `58b32a9` files (verified):

```
 visit tsd   code intended
   OL1   0  4.281    4.281
   OL2   0  9.223    9.223
   BD1   0  9.793    9.793
   BD2   0 10.187   10.187
   BD3   1  5.094    5.094
   BD4   2  1.273    2.547
   COd   0  4.281    4.281
   COp   4  0.268    0.268
```

Changing `t_half`, the path in `ondrug`, or the design reproduces the other entries of Tables 3 and 4. Applying either change of Section 9 before sourcing makes the two columns agree.

11. Limitations

- **One defect, one component.** The note treats only the off-drug mean of the drug response. The biomarker coupling rule, which drives the article's carryover power results, is a separate matter (Section 8.3).
- **Power check scope.** The paired simulation covers one design (Hybrid), one sample size ($N = 70$), one interaction strength ($c_{bm} = 0.25$), no censoring and 500 replicates per cell. It does not cover the article's full grid, its censoring patterns, or values of c_{bm} at which the published code repairs the correlation matrix.
- **The article's intent** is taken from its text (Section 3). The article does not give the carryover as an equation, so the anchored equation is our reading of that text; the authors may wish to confirm it.
- **Section 8.4** is inferred, not tested.

12. Reproducibility

Every number in Sections 1 and 4 to 10 that is labeled verified was computed by two scripts written for this note, in `analysis/scripts/carryover-recursion/` and run from the repository root:

- `53-carryover-examples.R`: Tables 2 to 5, both figures, the replica check against the published `generateData()`, the 40-case check of both fixes and of the current upstream file, and the re-exposure values.
- `53-ratio-coupling-check.R`: Table 8 (Section 9.5), the ratio coupling rule of the current upstream code under compounding and anchored means, for $s_f = 1, 2$ and $t_{1/2} = 1, 2$ weeks.
- `53-power-check.R`: the paired simulation of Section 8.3 (adapted from `analysis/scripts/quick-sim/hendrickson-p`

For example, `Rscript analysis/scripts/carryover-recursion/53-carryover-examples.R out53` writes the tables to `out53/` and the figures to `docs/figures/`. The scripts read the `58b32a9` sources from `analysis/data/quick-sim/hendrickson-58b32a9/R/`, which were checked to be byte-identical to the files fetched from GitHub at `58b32a9`, and the 2024 file from `analysis/scripts/quick-sim/hendrickson-original-comparison` which differs from upstream `06dac83` in one semantically identical line (a modern `data.table` column extraction). The parameter files `data/extracted_rp.rda` and `data/extracted_bp.rda` in this repository are identical to those at `58b32a9` (checked with `identical()`).

13. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013.
2. Hendrickson RC. pmsimstats: R package for simulation of aggregated N-of-1 trials. GitHub repository rchendrickson/pmsimstats, commit 58b32a92b99a8beb145de00c7bd8a0880f06f0f0 (2020-11-19); current `master` 06dac83f31ca85c994a6a2c1a0b98bc18ef233bd (2024-11-27). https://github.com/rchendrickson/pmsimstats
3. Venables WN, Ripley BD. *Modern Applied Statistics with S*. 4th ed. New York: Springer; 2002. doi:10.1007/978-0-387-21706-2.
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